Venkatesh K

Coimbatore, Tamil Nadu, India | +91 9025447567 | venkatesh.k21062005@gmail.com
linkedin.com/in/venkatesh-k-187448287 | github.com/venkatesh21bit

Education

Amrita Vishwa Vidyapeetham Coimbatore, India
B.Tech in Computer Science and Engineering ***Expected Graduation: May 2027***
CGPA: 7.8 / 10.0
Relevant Coursework: Design and Analysis of Algorithms, Neural Networks and Deep Learning, Machine Learning, Natural Language Processing, Linear Algebra & Optimization, Distributed Systems.

Research Interests

Large Language Model Optimization, Natural Language Understanding (NLU), Efficient Deep Learning, and Algorithmic Foundations of Machine Learning.

Publications

K Venkatesh, Dr. Uma Jothi, Harish J, Aishwarya Senthilkumar. *"A Hybrid Blockchain-IPFS Framework for Product Authentication and Supply Chain Management in E-Commerce."* Selected as one of the **Best Papers** at the 2nd International Conference on Grid and Parallel Computing (**ICGPC-2026**).

Research & Work Experience

Undergraduate Research Assistant Nov 2025 – Mar 2026
Amrita Vishwa Vidyapeetham Coimbatore, India

- Co-authored a peer-reviewed paper investigating cryptographic product verification and content-addressed deduplication layers across e-commerce frameworks.
- Designed and optimized automated scripts to encrypt product images via AES-256-GCM and manage SHA-256 metadata hashing on the Ethereum Sepolia testnet.
- Evaluated infrastructure performance, demonstrating a 97.9% reduction in on-chain storage and an average 86% infrastructure cost savings via scalable smart contracts.
- Formulated a WebSocket-based event-driven synchronization mechanism to broadcast real-time transaction updates with 100% verification success under controlled validation setups.

Edge AI Engineer Finalist Mar 2025 – Apr 2025
Sony Aitrios Hackathon Remote, India

- Engineered a computer vision pipeline targeted at automated warehouse logistics, specializing in real-time QR code localization, decoding, and physical package damage detection.
- Built and optimized a custom Machine Learning model utilizing localized structural features, achieving a 90% detection and classification accuracy on validation datasets.

NPM Package Developer Nov 2024 – Jan 2025
Winter of Code (WoC), Amrita ACM Club Coimbatore, India

- Contributed to the development of an open-source Command Line Interface (CLI) tool designed to bootstrap and generate optimized Express.js server architectures from modular, custom templates.
- Refactored core execution blocks, implemented strict automated test suites, and optimized package delivery mechanisms to reduce dependency weight and elevate execution reliability.

Projects

Predictive B2B Growth Engine Optimization SEED Business Challenge 2026

- Analyzed an enterprise dataset of 122,178 comprehensive business records across sales pipelines, multi-touch marketing attribution, and historical prospect engagement patterns.
- Formulated a data-driven diagnostic engine using predictive classification models to calculate Ideal Customer Profile (ICP) and alignment scores, isolating systemic funnel drop-offs.
- Structured an operational playbook and financial scaling simulation designed to reduce Customer Acquisition Cost (CAC) by 25-40%, boost lead conversion 2x-3x, and shrink the sales velocity cycle down to < 60 days.

QR Detection and Package Analytics Model Sony Aitrios Project

- Architected an end-to-end multi-task deep neural network tailored for resource-constrained edge computing modules, running localized object detection and sequence decoding.

- Implemented image transformation pipelines and bounding-box regressions to identify physical packaging deformations and extract structural identifiers concurrently.

etcd Leader Election Workflow Visualiser Distributed Systems Project

- Developed an interactive visualization system demonstrating real-time fault tolerance, node crash recovery, and state synchronization across a distributed 3-node etcd cluster using the Raft consensus protocol.
- Configured concurrent Python client engines competing for state-driven transactional locking via Compare-And-Swap (CAS) operations, implementing lease-based Time-To-Live (TTL) mechanics for failovers.
- Engineered a Flask API gateway utilizing Server-Sent Events (SSE) and persistent polling hooks to translate live PostgreSQL shared database mutations and cluster membership flags directly to a web interface.

Evaluation of Memory Caching Architectures Independent Research Project

- Formulated and analyzed implementations of Google Research's recent *"Memory Caching: RNNs with Growing Memory"* architecture to evaluate long-context understanding and retrieval properties.
- Profiled sub-quadratic sequence configurations by embedding Gated Residual Memory (GRM) components and Sparse Selective Caching (SSC) topologies onto Deep Linear Attention (DLA) systems.
- Tested Mixture-of-Experts (MoE) style token routers to contextually gate access to historical sequence checkpoints, successfully interpolating operational trade-offs between hidden state RNN and global attention bounds.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, SQL, Go, C/C++, Solidity
Frameworks & Deep Learning: PyTorch, PyTorch Geometric, JAX, Hugging Face, FastAPI, Flask, Express.js
Distributed Tools & Infra: etcd (Raft), PostgreSQL, IPFS, Docker, Git, Linux/Unix Shell, Ethereum Sepolia Testnet

Awards & Achievements

1st Place Winner, SEED Business Challenge 2026: Awarded a national cash prize of **35,000** for building a predictive enterprise growth diagnostic framework for NovaTech Solutions.

Best Paper Track Selection, ICGPC-2026: Awarded for novel hybrid ledger and off-chain storage optimization architectures designed for multi-platform e-commerce frameworks.

National Finalist, Sony Aitrios Hackathon (2025): Recognized among top edge-computing engineering submissions across India for structural vision analytics and real-time localized deployment.